

HAT-P-19b

R-1035

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-19_241207 · created 2026-07-30T07:05:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460651.725 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.027
Transit depth (Rp/Rs)^2	1.28 +/- 0.61 [%]
Orbital Inclination (inc)	89.63 +/- 1.18
Airmass coefficient 1 (a1)	0.944 +/- 0.023
Airmass coefficient 2 (a2)	0.052 +/- 0.015
Scatter in the residuals of the lightcurve fit is	2.46 %
Best Comparison Star	#1 - [481, 341]
Optimal Method	PSF photometry
Transit Duration (day)	0.1141 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.113 ± 0.027 vs 0.1418 (deviation 0.95σ)
Duration fit vs published	0.1141 d vs 0.1196 d (deviation 0.72σ)
Mid-time O-C	10.8 min
Depth SNR	3.7
Residual scatter	2.46 %

Light curve

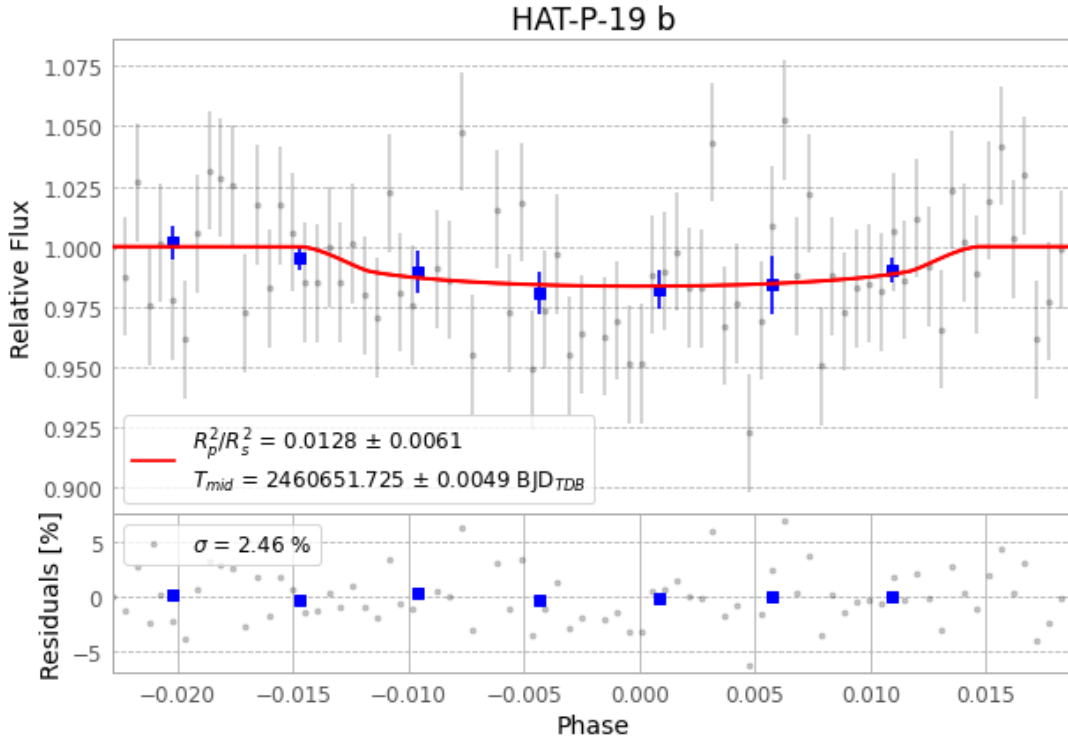


Figure 1 — FinalLightCurve_HAT-P-19 b_2024-12-06.png (SHA-256 65f729c19993c3e5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [393, 278], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c3c761c2bf2825f6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 77e40fd

Data provenance

81 FITS frames (53 MB), HATP-19241207030615.FITS → HATP-19241207070615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-19-241207.log (2881fa12785d...), FinalLightCurve_HAT-P-19 b_2024-12-06.png (65f729c19993...), FinalLightCurve_HAT-P-19 b_2024-12-06.csv (d430ed2f4a3e...)

Compute resources

Wall time 300.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 07:05Z.